$\eta = 1$ for one that holds one, and $\eta = 2$ for one excluded from the programme after failing Treatment First's condition (Section 4.9).

Each period the household chooses the bundle

$$\big(h,\, v,\, \tau,\, a'\big) \in \{h_1, h_2, h_3, h_4\} \times \mathbb{R}_+ \times \{0, 1\} \times \mathbb{R}_+, \tag{10}$$

where h is the dwelling, of which h_1 is a shelter, v is drug consumption, τ is whether or not the household is in treatment for the SUD, available only to a household that has the disorder, and a' is its next period's assets. Labour is inelastically supplied when employed.

4.2 Preferences

Flow utility is the stylized model's, with housing in the bundle and the child present for the first J_K years. Consumption c and housing h are shared through the equivalence scale $\xi(j)$, which counts two members while the child is at home and one after. The drug enters the parent's utility alone, so the parent and the child hold the bundles

$$X_P = \Big[\Big(\frac{c}{\xi(j)}\Big)^{1-\gamma}\Big(\frac{h}{\xi(j)}\Big)^{\gamma}\Big]^{1-\theta}(b+v)^{\theta}, \qquad X_c = \Big[\Big(\frac{c}{\xi(j)}\Big)^{1-\gamma}\Big(\frac{h}{\xi(j)}\Big)^{\gamma}\Big]^{1-\theta} b^{\theta}, \tag{11}$$

and period utility is

$$u\big(c, h, v, \tau\big) \;=\; U(X_P) \;+\; \omega_c\,(\xi(j)-1)\,U(X_c) \;+\; \aleph \;-\; \tau\,\bar{\tau}^{u}, \qquad U(X) = \frac{X^{1-\sigma}}{1-\sigma}. \tag{12}$$

γ is housing's share of the non-drug bundle and θ the drug's share of the whole, σ is the coefficient of relative risk aversion, $\aleph$ is a flow value of being alive, counted once per household, and $\bar{\tau}^{u}$ is the utility cost of a year in treatment, which is described in Section 4.6. The child's term is its own flow utility, counted as $\xi(j) - 1$ of the household and weighted by the parent at ω_c. The child gets none of the drug, so a parent's drug spending reaches it through a reduction in the consumption and housing they share.

4.3 Income, taxes and transfers

Productivity. Log productivity follows a persistent process over working life, $\log y_{j+1} = \rho_y \log y_j + \zeta_{j+1}$ with $\zeta \sim \mathcal{N}(0, \sigma_\zeta^2)$, and is held fixed at its last working value in retirement. When employed, a household earns $y\,\varepsilon(j)$. The term $\varepsilon(j)$ is a deterministic age profile, and y is normalised such that it averages one upon entry. A household inherits its productivity at entry from its parent, and a child who enters with the penalty keeps a lower productivity throughout their life. I explain the intergenerational transmission in detail in Section 4.8.

Before-tax income. Before-tax income depends on age and employment,

$$\tilde{y}(j, y, e) = \begin{cases} y\,\varepsilon(j) & j < J_R,\ e = 1 \quad \text{(employed)} \\ \psi_{ui}\,\min\{y\,\varepsilon(j),\ \bar{y}\} & j < J_R,\ e = 2 \quad \text{(insured, first year out of work)} \\ 0 & j < J_R,\ e = 3 \quad \text{(out of work, uninsured)} \\ S(y) & j \geq J_R \quad \text{(retired)}. \end{cases} \tag{13}$$

Unemployment insurance (UI) replaces a share ψ_{ui} of earnings on a base capped at $\bar{y}$, and it is paid only to a share p_{ui} of those losing their job, mimicking the share of insured separations. After the first year, unemployed households, insured and uninsured alike, receive zero UI ($e = 3$). Social security $S(y)$ follows the statutory formula.[16]

[16] Let $y^s = \min\{y,\, y^{\max}\}$ denote the household's productivity at its last working age up to the taxable maximum $y^{\max}$, the ceiling above which earnings no longer raise the benefit. Then social security payments are $S(y) = \bar{w}\max\big\{\underline{S},\ \alpha_1 \min\{y^s, k_1\} + \alpha_2 \max\{0, \min\{y^s, k_2\} - k_1\} + \alpha_3 \max\{0, y^s - k_2\}\big\}$, where $k_1 < k_2$ are the two bend points, $\alpha_1 > \alpha_2 > \alpha_3$ the shares replaced below, between and above them, $\underline{S}$ the Supplemental Security Income (SSI) payment, and $\bar{w}$ the benchmark's average labour income. Their values are given in Appendix A.

The SUD and SMI act on whether the household is employed, through the employment transitions below. There is no disability insurance in the model, so SMI brings no income of its own.[17]

Taxes. Before-tax income is taxed with the Heathcote et al. (2017) function and capital income at a flat rate, so that after-tax income is

$$y^n = \min\left\{\tilde{y},\ \tau_\ell\,\tilde{y}^{\varphi}\right\}\ +\ (1-\tau_k)\,r\,a, \tag{14}$$

where τ_ℓ scales after-tax income and so sets the level of the tax, which falls as τ_ℓ rises, and $1-\varphi$ sets its progressivity. The flat tax rate on capital income is τ_k, and r is the exogenous interest rate on savings.[18]

Transfers. Food assistance pays the maximum allotment $\underline{c}$ less the share ϱ of income. The voucher pays a holder that is eligible this period the part of its rent above a tenant contribution of κ_r of income, up to the fair market rent (FMR),

$$\begin{aligned} F &= \max\left\{0,\ \underline{c}-\varrho\,\hat{y}\right\},\\ Vch &= \mathbf{1}\{\eta=1\}\,\mathbf{1}\{\hat{y}<\kappa_V y^{\text{med}}\}\,\mathbf{1}\{h\neq h_1\}\,\min\Big\{\max\{0,\ p_r h-\kappa_r\hat{y}\},\ \max\{0,\ FMR-\kappa_r\hat{y}\}\Big\}, \end{aligned} \tag{15}$$

where $\hat{y}=\tilde{y}+ra$ is the income both programmes are means-tested on, κ_V and median income y^{med} are the eligibility rule in Section 4.9, and FMR is the average rent paid by housed households in the benchmark. A voucher holder in a shelter keeps the voucher and is paid nothing until it rents again, as long as it keeps its eligibility.

Each entering household receives a one-time endowment T_0 from bequests left by deceased households, and the government collects the rest of the estates of the dead.

4.4 The budget constraint

The household's budget constraint is as follows:

$$c+p_v v+a' \ =\ y^n+a+T_0\mathbf{1}\{j=1\}\ -\ p_r\,h\,\mathbf{1}\{h\neq h_1\}\ +\ Vch+F, \qquad c\geq F,\ v\geq 0,\ a'\geq 0. \tag{16}$$

Food assistance is paid in kind, so it cannot directly fund the drug, the rent or saving. The drug price is p_v, which is endogenous, whereas the rent per unit of housing p_r is fixed and held at its benchmark value in every simulation.[19] Treatment costs the household utility only; the government pays for it (see Section 4.11).

4.5 Housing and the homelessness loop

Housing is the four-point grid of (10): h_1 is literal homelessness, sleeping rough or in an emergency shelter, which is free, and $h_2<h_3<h_4$ are three rental sizes. Homelessness raises mortality (20), worsens both employment margins (21) and lowers the chance of recovering from the SUD (18), but does not affect SMI (Section 4.7).

4.6 Drug use, the SUD, recovery and treatment

A household that does not have the disorder, either $d=0$ (never had it) or $d=1$ (had it but recovered), acquires it ($d=2$) with probability π_d if it uses,[20]

$$\Pr(d'=2\mid d\neq 2,\,v)=\pi_d\,\mathbf{1}\{v>0\}, \tag{17}$$

[17] In the data only a minority of adults with SMI receive disability benefits on the basis of a mental disorder: about 3 million, through Social Security Disability Insurance or SSI, against 14.6 million adults with SMI in NSDUH 2023, roughly one in five.

[18] The minimum keeps the labour income tax from turning negative, since at low incomes $\tau_\ell\tilde{y}^{\varphi}$ exceeds $\tilde{y}$ itself.

[19] In earlier versions of this work, I solved the model with endogenous labour supply, rents, homeownership and house prices. These margins turned out to have a negligible effect on the results, so for conciseness I removed them and hold the rent fixed.

[20] In earlier versions of the model, I allowed the probability of developing a SUD to rise with use, but since it had little impact on the results, I removed it for simplicity.

and a household that has recovered can acquire the disorder again. A household may begin using only up to an age J_E.[21] A household with the SUD recovers, to $d = 1$, with probability

$$\pi_r(h, m, \tau) = \min\Big\{1,\ \bar{\pi}_r \left(\mathbf{1}\{h = h_1\}\,\mu_H + \mathbf{1}\{h \neq h_1\}\right)\left(\mathbf{1}\{m = 1\}\,\mu_m + \mathbf{1}\{m = 0\}\right) \times \left(\mathbf{1}\{\tau = 1\}\,g + \mathbf{1}\{\tau = 0\}\right)\Big\}, \tag{18}$$

where $\bar{\pi}_r$ is the baseline annual chance of recovering while housed, without SMI and untreated. Three multipliers affect it. Recovery is harder when homeless ($\mu_H < 1$) as well as with SMI ($\mu_m < 1$), but is more likely in treatment ($g > 1$).

Households who go to treatment τ pay a utility cost $\bar{\tau}^u$, shown in (12). What they gain is a higher chance of recovery, with $g > 1$. Treatment does not require a household to stop using, but as long as the disorder remains, it still raises the risk of mortality.

4.7 SMI

m is a two-state Markov chain representing SMI, mental illness severe enough to cause substantial functional impairment. It affects a small share of adults, whereas any mental illness affects roughly a quarter of adults and includes mild anxiety. Onset follows an age profile $\varsigma_m(j)$, normalised to one at age 26 and scaled by the level ω_m, and a household that carries the SUD is $\iota > 1$ times more likely to develop SMI than one that does not,

$$\Pr(m' = 1 \mid m = 0, d) = \min\left\{1,\ \omega_m\,\varsigma_m(j)\left(\mathbf{1}\{d = 2\}\,\iota + \mathbf{1}\{d \neq 2\}\right)\right\}, \qquad \Pr(m' = 0 \mid m = 1) = \lambda_m. \tag{19}$$

A household with SMI recovers from it with the probability λ_m each year, whatever its housing and whether or not it carries the SUD.

SMI then affects three objects: the chance of finding a job when unemployed, through (21); mortality, through (20), both in Section 4.8; and the SUD recovery hazard, through μ_m in (18) in Section 4.6. It does not enter the utility function.

The SMI chain does not depend on housing, because in the At Home / Chez Soi randomised trial, housing had no detectable effect on the symptoms of homeless adults with SMI (Stergiopoulos et al., 2015).

4.8 What the SUD, SMI, and homelessness cost

Death risk. Write $1 - \bar{\phi}(j)$ for the probability of dying at age j of a household that is housed and has neither the disorder nor SMI; the three multipliers below raise it for every other household. Write $q^0(j,h) = \min\{1, \left(\mathbf{1}\{h = h_1\}\,\kappa_{\phi H} + \mathbf{1}\{h \neq h_1\}\right)(1 - \bar{\phi}(j))\}$ and $q^1(j,h) = \min\{1, \kappa_{\phi d}\left(\mathbf{1}\{h = h_1\}\,\kappa_{\phi H} + \mathbf{1}\{h \neq h_1\}\right)(1 - \bar{\phi}(j))\}$ for the probability of dying without and with the disorder, respectively. In the stylized model $\kappa_{\phi d}$ was an exponent on the survival probability; here it multiplies the death hazard. The probability of dying is then

$$1 - \phi(j, h, d, m) = \min\Big\{1,\ \left(\mathbf{1}\{m = 1\}\,\kappa_{\phi m} + \mathbf{1}\{m = 0\}\right) q^0(j,h) + \mathbf{1}\{d = 2\}\left[q^1(j,h) - q^0(j,h)\right]\Big\}, \tag{20}$$

so $\kappa_{\phi H} > 1$ multiplies the death probability when a household is in a shelter, $\kappa_{\phi d} > 1$ when a household has a SUD, and $\kappa_{\phi m} > 1$ the baseline risk when a household has SMI, to which the excess risk of the disorder is added.

[21] Without this limit, using at old ages has little-to-no consequence: too little life remains for the disorder to damage, so the model produces too much old-age use. The cause is that death is certain at the last age J, so close to it a household knows how little life remains, whereas in real life the vast majority of people do not know exactly when they will die. Its value is in Appendix A. It stops only households without a SUD from starting to use; a household that already has a SUD at that age keeps using.

Unemployment risk. A job is lost with probability π^{eu} and found with probability π^{ue},

$$\begin{aligned}\pi^{eu}(y,h) &= \min\Big\{1,\ \bar{\pi}^{eu}\,\varpi(y)\,\big(1+\mathbf{1}\{h=h_1\}\,\kappa_H\big)\Big\},\\ \pi^{ue}(h,d',m') &= \min\Big\{1,\ \frac{\bar{\pi}^{ue}\,\big(1-\mathbf{1}\{h=h_1\}\,\kappa_H\big)}{\big(\mathbf{1}\{d'=2\}\,\kappa_d^{ue}+\mathbf{1}\{d'\neq 2\}\big)\big(\mathbf{1}\{m'=1\}\,\kappa_m^{ue}+\mathbf{1}\{m'=0\}\big)}\Big\},\end{aligned} \tag{21}$$

with $\bar{\pi}^{eu}$ and $\bar{\pi}^{ue}$ the baseline separation and job-finding rates, $\varpi(y)$ a factor that makes separations likelier at low productivity, and the divisors κ_d^{ue} and κ_m^{ue} making a job harder to find with the SUD and with SMI. Neither the SUD nor SMI raises separations; each only reduces the chance of regaining employment. κ_H does both for a household in a shelter.

The child's productivity penalty. A child whose parent had the disorder at any point while the child lived at home, before J_K or before the parent died if that came first, starts adult life with its productivity lowered by the proportion δ_d in (22) below, and the penalty stays with the child for life.

A child's log productivity at entry is drawn around its parent's, with persistence ρ. The innovation keeps the spread of productivity at entry the same in every generation. If the parent had the disorder at any point before the child became an adult, the penalty scales the child's productivity down for life:

$$\begin{aligned}\log y_1^{\text{child}} &= \rho\,\log y^{\text{parent}} + (1-\rho)\,\mu + \epsilon\\ &\quad + \log(1-\delta_d)\,\mathbf{1}\{\text{parent has } d\geq 1 \text{ when its child becomes an adult}\},\end{aligned} \tag{22}$$

with y_1^{child} the child's productivity in the period it becomes an adult in the workforce ($j=1$), y^{parent} the parent's productivity in the period just before its child becomes an adult, μ the mean of log productivity at entry and ϵ normally distributed with mean zero and variance $1-\rho^2$ times the variance of log productivity at entry. The penalised child's productivity is lower, which means that its own child is also more likely to inherit a lower draw. Through this persistence, the penalty ripples through the whole lineage, and not just the child whose parent had a SUD during their childhood.

4.9 The voucher and the two policies

A household is eligible for a voucher while its income is below the share κ_V of median income,

$$\hat{y} \;<\; \kappa_V\, y^{\text{med}}, \tag{23}$$

with $\hat{y}$ the income of (15). A holder keeps its voucher while it is eligible, and an eligible household without one draws one with probability π_V. Under Housing First that is the whole rule.

Under Treatment First the programme is conditional. A household fails the condition if it uses, $v>0$, or has the disorder and is not in treatment, $d=2$ and $\tau=0$. A household with a SUD must be in treatment (paid for by the government), as the executive order directs, and no household may use even while in treatment, as the guidance implementing the order requires of drug-free housing.[22]

Income-eligible holders and income-eligible households that draw a voucher are tested, at a cost c_{test} each to the government, and one that fails the condition is detected with probability p_{det}. This means that enforcement is not perfect: with probability $1-p_{\text{det}}>0$ a household that fails the condition actually gets to keep its voucher. A detected holder keeps this period's payment and loses the voucher from the next period; a detected applicant never receives it. In both cases, the household is excluded from the voucher programme for a period of time ($\eta=2$): it cannot draw a voucher whatever its income, and it returns to the lottery ($\eta=0$) with probability ψ_x each year.

[22] Executive Order 14321, Section 5(b); the guidance is *Addressing Homelessness and Addiction through "Treatment First"* (2026), p. 9, issued by the Office of National Drug Control Policy, the Department of Health and Human Services and HUD. The Housing Choice Voucher programme bars admission for current illegal drug use (Code of Federal Regulations (CFR), Title 24, Section 982.553) but does not test; Housing First here is a programme that does not test. The order also encourages the civil commitment of people with mental illness who pose a risk to themselves or others, or who live on the street and cannot care for themselves. In the model, treatment is always the household's choice: a household with a SUD that declines it loses its voucher if the screen catches it.

None of p_{det}, ψ_x and c_{test} binds in the benchmark, where no voucher is conditional, so all three are set with the experiment in Section 6 rather than in the calibration.

4.10 The household's dynamic problem

Write $\Gamma\big(s' \mid s,\, h, v, \tau, a'\big)$ for the law of motion of the state that the transitions above define, given the state and the choices. Then, for $j = 1, \ldots, J$ and with $V(J{+}1, \cdot) = 0$,

$$
\begin{aligned}
V(j,s) = \max_{\{h,\,v,\,\tau,\,a'\}} \Bigg\{ & U(X_P) + \omega_c(\xi(j)-1)\,U(X_c) + \aleph - \tau\bar{\tau}^{u} \\
& +\ \beta\,\phi(j,h,d,m) \int_{\mathcal{S}} V(j{+}1, s')\,\Gamma\big(\mathrm{d}s' \mid s, h, v, \tau, a'\big) \\
& -\ \beta\,\chi(j)\,\omega_c\,D(y,d) \Bigg\}, \qquad \chi(j) = \begin{cases} 1-\phi(j,h,d,m) & j < J_K \\ 1 & j = J_K \\ 0 & j > J_K, \end{cases}
\end{aligned}
\tag{24}
$$

subject to the budget (16), the income and transfer rules (13) to (15) in Section 4.3, and the restrictions that treatment is available only to a household with $d = 2$ and that no household begins using after J_E.

The first line is the flow of (12). The second is the household's own continuation. The third is the impact of the parent on the child's adult value, counted only when the child leaves: at J_K with certainty, and before then with the probability that the parent dies first. $D(y,d)$ is what the parent's disorder costs the child, in the child's own lifetime value,

$$
D(y,d) = \mathbb{E}\big[V(1, s_{\text{child}}) \,\big|\, y,\ 0\big] - \mathbb{E}\big[V(1, s_{\text{child}}) \,\big|\, y,\ d\big] \ \geq\ 0, \tag{25}
$$

where the child is valued as it enters at $j = 1$ with the initial states described in Section 4.13, and its productivity is drawn as in (22) in Section 4.8 from a parent with productivity y. In the first term of (25) the parent has never had the disorder, and in the second it has its actual d in the period the child leaves. If it never had a SUD, there is no charge to the parent ($D(y,0) = 0$); if it had a SUD before the child leaves ($d \geq 1$), it is charged ($D(y,d) > 0$).[23]

One weight ω_c multiplies both child terms, the flow of childhood and the value of the child's adulthood. At $\omega_c = 0$ the damage a household does to its child is entirely external; at $\omega_c > 0$ it internalises the share ω_c and only $1 - \omega_c$ remains external.

The policy functions $h(j,s)$, $v(j,s)$, $\tau(j,s)$ and $a'(j,s)$ are the dwelling, drug use, treatment and saving that solve (24) at age j and state s.

4.11 The government

Write $\Phi(j,s)$ for the distribution of the households of age j over the states s; its law of motion is in Section 4.13. The government's budget constraint is

$$
\begin{aligned}
\mathcal{R} \ = & \sum_{j=1}^{J} \int_{\mathcal{S}} \Big[\, \tilde{y}(j,y,e) - \min\big\{\tilde{y}(j,y,e),\ \tau_\ell\,\tilde{y}(j,y,e)^{\varphi}\big\} + \tau_k\, r\, a \\
& \qquad + \big(1 - \phi\big(j, h(j,s), d, m\big)\big)\, a'(j,s) \Big]\, \mathrm{d}\Phi(j,s) \ -\ T_0 \\
& - \sum_{j=1}^{J} \int_{\mathcal{S}} \Big[\, UI(j,y,e) + S(j,y) + F(j,s) + Vch(j,s) + c_{\text{test}}\,\mathbf{1}^{\text{test}}(j,s) + c_{\text{treat}}\,\mathbf{1}\{d=2,\ \tau(j,s)=1\} \Big]\, \mathrm{d}\Phi(j,s).
\end{aligned}
\tag{26}
$$

[23] The term is a difference and not a level. Written as a level, ω_c times the child's value, the term would be worth more to a household the likelier the household is to die, because the death branch carries it. It would then pay most to the households with a SUD, for reasons that have nothing to do with the child. As a difference, the term is zero for a parent who has never had the disorder and a cost otherwise.

The first line is revenue from the labour and capital income taxes of (14). The second is the estates of the households who die, which the government collects, less the endowment T_0 it gives the unit mass of entrants. The third is what the government pays: UI, $UI(j,y,e) = \tilde{y}(j,y,e)\,\mathbf{1}\{j < J_R,\, e = 2\}$; social security, $S(j,y) = S(y)\,\mathbf{1}\{j \geq J_R\}$; food assistance $F(j,s)$ and the voucher $Vch(j,s)$ of (15), at the household's dwelling $h(j,s)$; under Treatment First, the cost c_{test} of each household tested, $\mathbf{1}^{\text{test}}(j,s) = 1$; and, under both policies, the cost c_{treat} of a year of treatment for each household with the SUD that takes it up, $\tau(j,s) = 1$, since treatment is publicly funded.

In the benchmark the tax parameters are set as in Section 5 and $\mathcal{R}$ is whatever surplus they produce. In every experiment the taxes are held at their benchmark values, and the probability of drawing a voucher π_V adjusts so that $\mathcal{R}$ equals its benchmark value $\bar{\mathcal{R}}$, unless the experiment fixes π_V.

4.12 The drug market

The price of the drug clears its market,

$$p_v = \nu_1 S_v^{1/\varepsilon_s}, \qquad S_v = \sum_{j=1}^{J} \int v(j,s)\,\mathrm{d}\Phi(j,s), \tag{27}$$

where ε_s is the price elasticity of supply and ν_1 is normalised so that the price is one in the benchmark, and then held fixed across experiments.[24]

4.13 The law of motion

Given the policy functions, the distribution over states evolves by pushing households through their choices and then through the shocks,

$$\Phi(j{+}1, \mathcal{S}') \;=\; \int_{\mathcal{S}} \phi\big(j, h(j,s), d, m\big)\, \Gamma\big(\mathcal{S}' \mid s,\, h(j,s), v(j,s), \tau(j,s), a'(j,s)\big)\,\mathrm{d}\Phi(j,s), \tag{28}$$

for $j = 1, \ldots, J-1$ and every measurable set $\mathcal{S}'$ of the state space $\mathcal{S}$ of (9), with the integral over the state s of the households of age j, together with the inflow of entrants

$$\Phi(1, \mathcal{S}') \;=\; \frac{\sum_{j \leq J_K} \int_{\mathcal{S}} \chi(j)\, \mathcal{N}\big(\mathcal{S}' \mid s\big)\,\mathrm{d}\Phi(j,s)}{\sum_{j \leq J_K} \int_{\mathcal{S}} \chi(j)\,\mathrm{d}\Phi(j,s)}, \tag{29}$$

where the integrals are again over s, $\mathcal{N}$ places the child according to (22) and $\chi(j)$ is the weight defined in Section 4.10. An entrant starts with the endowment T_0, is employed, has never had the disorder, holds no voucher, and carries SMI with the prevalence observed at age 26.[25]

4.14 Competitive equilibrium

A competitive equilibrium is a collection of

- value functions $\big\{\{V_t(j,\cdot)\}_{j=1}^{J}\big\}_{t=0}^{\infty}$;
- policy functions $\big\{\{h_t(j,\cdot),\, v_t(j,\cdot),\, \tau_t(j,\cdot),\, a'_t(j,\cdot)\}_{j=1}^{J}\big\}_{t=0}^{\infty}$;
- distributions $\big\{\{\Phi_t(j,\cdot)\}_{j=1}^{J}\big\}_{t=0}^{\infty}$; and
- prices, policies and aggregates $\big\{p_{vt},\, \pi_{Vt}\big\}_{t=0}^{\infty}$, together with $\bar{y}$, y^{med} and FMR, which carry no date because they are kept fixed at their benchmark values,

such that, at every date t,

[24] The suppliers are outside the economy: what they earn along the supply curve is counted neither in welfare nor in the budget.

[25] A disorder acquired before that age is therefore acquired in the model only after entry; however, the share of adults with a SUD still matches the data (Section 5).

1. the policy functions solve (24) given its terminal condition, and $\{V_t(j,\cdot)\}_{j=1}^{J}$ are the associated values;
2. $\Phi_{t+1}(j{+}1,\cdot)$ follows from $\Phi_t(j,\cdot)$ by the law of motion (28) together with the entrants of (29) and (22);
3. the government budget (26) holds at π_{Vt} with $\mathcal{R} = \bar{\mathcal{R}}$;
4. the drug market clears at p_{vt}, so (27) holds.

A stationary equilibrium is a special case in which all objects are constant. I calibrate the model in a stationary equilibrium, and the long-run policy comparisons also report stationary equilibria. When I hold π_V or the drug price at their benchmark values in any experiment, I exclude the budget condition or the drug-market condition, respectively.

5 Calibrating the quantitative model

The model has fifty-nine parameters, some of which vary by age. Thirty-eight are calibrated externally, from data or the literature, and twenty-one internally. The twenty-six that carry this paper's story are in Table 3, with the moment, source, data and model value behind each, and are discussed below in numbered blocks of parameters that work through similar mechanisms. The other thirty-three, common to most life-cycle models, are in Appendix A to save the reader time; its blocks continue the numbering. Three further parameters govern drug testing and exclusion from the voucher; they play no role in the benchmark and are set with the experiment in Section 6. `data/DATA_APPENDIX.md` in the replication package documents the dataset, year, universe and computation behind every estimate, and the package carries all datasets, so nothing has to be downloaded.

5.1 The twenty-six parameters that carry the story of this paper

Table 3: The twenty-six parameters that carry the story of this paper

Parameter	Description	Set	Moment and source	Data	Model	Value
Block 1: Who uses drugs						
θ	Weight on drugs in utility	Int.	Drug spending over income, Midgette et al. (2019)	16.0%	14.8%	0.165
b	Marginal utility from the first unit of drugs	Int.	Any drug use disorder, NSDUH 2024	8.90%	8.96%	0.0054
π_d	Chance that a year of use leads to a SUD	Int.	Share of users with a SUD, NSDUH 2024	41.3%	42.6%	0.0345
$\kappa_{\phi d}$	Increased mortality with a SUD	Int.	Income of drug users over average, NSDUH 2024	74%	62%	4.00
$\aleph$	Flow value of being alive	Int.	VSL, Viscusi and Aldy (2003)	80	81	87.6
Block 2: Costs of homelessness and vouchers						
κ_d^{ue}	How much less likely it is to regain employment with a SUD	Int.	SUD share of the homeless, Gutwinski et al. (2021)	21.7%	20.2%	3.78
h_1	Service flow of homelessness	Int.	Homelessness band, HUD PiT 2023 and Burt et al. (2001)	0.40–1.00%	0.62%	0.0216
$\kappa_{\phi H}$	Mortality when homeless	Ext.	Meyer et al. (2023)	1.60	n.a.	1.60
κ_H	Employment penalty of homelessness	Ext.	Desmond and Gershenson (2016)	0.20	n.a.	0.20
$\underline{c}$	Food stamp maximum allotment	Int.	SNAP outlays over output	0.42%	0.43%	0.0879
π_V	Chance of drawing a voucher	Int.	Recipients over eligible, HUD 2023	26.6%	27.5%	0.0675
T_0	Endowment at entry to adulthood ($j = 1$)	Int.	Share of homeless aged 25 to 34, HUD PiT 2024	26.0%	26.4%	0.111
Block 3: SMI chain						
ι	SUD multiplier on SMI onset	Ext.	Danish registers, Nielsen et al. (2017)	2.00	n.a.	2.00
λ_m	Annual chance of recovering from SMI	Ext.	SMI ten-year recovery rate, Åsbø et al. (2022)	50%	n.a.	0.066
ω_m	Level of the SMI onset profile	Int.	SMI share of the population, NSDUH 2024	4.82%	4.83%	0.0037
κ_m^{ue}	How much less likely it is to regain employment with SMI	Int.	SMI share of the homeless, Gutwinski et al. (2021)	25%	24.7%	7.71
μ_m	SUD recovery multiplier for SMI	Ext.	Dual-diagnosis relapse, Bradizza et al. (2006)	0.82	n.a.	0.82
$\kappa_{\phi m}$	Mortality multiplier for SMI	Ext.	Olfson et al. (2015)	2.50	n.a.	2.50
Block 4: Recovery and treatment						
$\bar{\pi}_r$	Baseline SUD recovery rate	Ext.	SUD ten-year remission rate, Vaillant (1995)	22%	n.a.	0.025
μ_H	SUD recovery when homeless	Int.	Abstinence, housed/homeless, Milby et al. (2005)	1.47	1.45	0.696
g	Treatment effect on recovery	Ext.	Remission on buprenorphine, Paul et al. (2024)	2.00	n.a.	2.00
$\bar{\tau}^u$	Utility cost of treatment	Int.	Treatment take-up, NSDUH 2024	24.5%	24.9%	4.49
c_{treat}	Public cost of a year of treatment	Ext.	Residential treatment episode, Roebuck et al. (2003)	$9,426	n.a.	0.133
Block 5: Externalities						
ω_c	Parent's weight on its child	Int.	Two-adult over lone-parent households' use, less one, NSDUH 2024	17.5%	17.7%	0.085
δ_d	Earnings the child loses because of a parent's SUD	Ext.	Earnings lost in utero, Nilsson (2017)	0.24	n.a.	0.24
Block 6: Drug market						
ε_s	Price elasticity of drug supply	Ext.	Mace et al. (2020)	1.46	n.a.	1.46

Notes: the Model column is the Housing First benchmark. n.a. marks a parameter set from outside the model, which has no moment of its own to reproduce.

Block 1: Who uses drugs. The five parameters in this block set participation and, more importantly, *who* participates. They decide how much there is for a policy to affect.

θ, **the weight on drugs in utility.** The weight in utility given to drugs mostly determines what share of its budget a household that uses drugs spends on drugs. To a lesser extent, a higher weight also makes using at all more attractive. As it rises, use therefore rises on both the extensive and the intensive margin, and with it the externalities and what either policy has to act on. It is calibrated internally, at 0.17, to aggregate drug spending by users over their aggregate income, 16.0 percent, from retail drug spending in Midgette et al. (2019) and, in NSDUH 2024, the incomes of adults aged 26 and over who use drugs.

b, **the marginal utility from the first unit of drugs.** The marginal value of the first unit of drugs goes as θ/b, so a larger b makes the first units worth less and a household with little to spend buys none. As b falls, the marginal utility from the first unit rises, and so do overall drug use and the externalities that policy can affect. It targets a participation rate of 8.90 percent, the NSDUH 2024 rate of a drug use disorder of any severity among adults aged 26 and over.[26]

π_d, **the chance that a year of use leads to a SUD.** By governing how likely drug use is to become a disorder, this parameter determines how risky drug use is. When it rises, the externalities and moral hazard grow. It targets the share of users who have a SUD, 41.3 percent among adults aged 26 and over in NSDUH 2024.

$\kappa_{\phi d}$, **increased mortality with a SUD.** $\kappa_{\phi d}$ sets how much a SUD raises the chance of dying, and mortality truncates a user's value of life. Because this is an uninsurable risk, a rise in $\kappa_{\phi d}$ hits the households with the highest values of life hardest, which reverses the income gradient; as the stylized model of Section 3 shows, this makes it a key parameter. It is calibrated internally, at 4.00, to the income of users relative to the average, a value within the range of the mortality rate ratios of a SUD in the literature: 3.91 in the Danish registers (Plana-Ripoll et al., 2019) and 6.6 among Norwegians in specialist care for a SUD (Heiberg et al., 2018). Users earn 62 percent of the average in the model against 74 percent in the data.

$\aleph$, **the flow value of being alive.** While $\kappa_{\phi d}$ sets how much a SUD raises the chance of dying, $\aleph$ decides the value of life. As it rises, dying costs more and use falls. It targets a VSL of 80 times average labour income (Viscusi and Aldy, 2003).[27]

Block 2: Costs of homelessness and vouchers. Both policies work on this block, from opposite ends: each parameter says how welfare reducing a state is, and so what a voucher and its removal are worth in it.

κ_d^{ue}, **how much less likely it is to regain employment with a SUD.** An unemployed household with a SUD finds work at the rate $\bar{\pi}^{ue}$ divided by $\kappa_d^{ue} > 1$. As κ_d^{ue} rises, the expected unemployment spells of households with a SUD lengthen and they are likelier to lose their housing: this is the link from the disorder to the street, and it shapes how much Treatment First deters. It is calibrated internally to the SUD share of the homeless, 21.7 percent in Gutwinski et al. (2021).

h_1, **the service flow of homelessness.** This parameter sets how welfare reducing the outside option is, which affects the value of life: a household with little to lose from dying faces a smaller mortality cost of using. As h_1 rises, a shelter is less welfare reducing, a voucher is worth less and Treatment First has less to act on. It targets a homelessness rate within 0.40 to 1.00 percent of households: the floor is HUD's 2023 PiT count, which counts a single night; the ceiling is the share homeless at some point over a year, from Burt et al. (2001). Its value is 53 percent of the smallest rental (Appendix A), and the model delivers an aggregate homelessness rate of 0.62 percent.

$\kappa_{\phi H}$, **mortality when homeless.** This parameter sets how much more likely a homeless household is to die. As it rises, the cost of homelessness rises, a voucher is worth more and Treatment First's deterrence is stronger. It is set externally to multiply baseline mortality by 1.60 at every age, the mortality ratio for homeless adults against poor housed adults in Meyer et al. (2023).

[26] A user in the model is therefore a household whose use of any illicit drug, marijuana among them, amounts to a disorder, mild included. A household with a SUD is one whose disorder is moderate or severe (next item). Use that never reaches a disorder is not modelled.

[27] No household has a negative value of life in equilibrium, so death is never optimal. Because the value of being alive enters utility additively and utility is concave, the VSL rises faster than consumption, roughly with its square at the model's risk aversion, as in Hall and Jones (2007). Estimates of the income elasticity of the VSL are lower across people at a point in time, 0.5 to 0.6 in Viscusi and Aldy (2003), than over time, 1.5 to 1.7 in Costa and Kahn (2004).

κ_H, **the employment penalty of homelessness.** This parameter sets how much likelier a household in a shelter is to lose its job and how much less likely it is to find one: separations are $1 + \kappa_H$ times as frequent and job finding $1 - \kappa_H$ times as fast. Housing First recovers that employment directly, whereas Treatment First does so only as the fall in SUD lowers unemployment and homelessness over time. As κ_H rises, homelessness costs more, and both policies have more to act on. It is set externally at 0.20, from Desmond and Gershenson (2016).

$\underline{c}$, **the food stamp maximum allotment.** This parameter sets how low consumption can go for the poorest households, with no income, assets or voucher, and so where utility is bounded from below. As it rises, the value of being alive rises, mortality costs more and Treatment First has more to deter. It is calibrated internally to food stamp spending over output, 0.42 percent: outlays from the United States Department of Agriculture (USDA) over GDP.

π_V, **the chance of drawing a voucher.** This is the policy dial itself, and it holds the whole trade-off. Raising it houses more households, which reduces the *childhood homelessness externality*, and makes life worth more, which makes Treatment First's deterrence more effective. In the benchmark it is calibrated internally, at 0.068, so that the share of income-eligible households who hold a voucher is aligned with the 26.6 percent in HUD's Picture of Subsidized Households, 2023.

T_0, **the endowment at entry to adulthood** ($j = 1$)**.** A child who becomes an adult starts adult life with T_0 times average earnings, which sets how likely a young adult is to end up in a shelter in its first years. As it rises, fewer parents are homeless, and Treatment First has less of an effect on drug use during parenthood, and thus less of an effect on children. Its moment is the share of the homeless aged 25 to 34, 26.0 percent in HUD's 2024 PiT count.

Block 3: SMI chain. This block determines how much beyond SUDs Treatment First prevents. This is because a SUD raises the chance of SMI, and SMI lowers employment and raises mortality.

ι, **the SUD multiplier on SMI onset.** This is the key parameter of this block because it determines how much a SUD affects the onset of SMI. As it rises, a SUD increases the prevalence of SMI, and Treatment First reaches further, because preventing the disorder then also lowers overall SMI. It is set externally at 2.00, between the hazard ratio of 1.34 for any SUD that Hjorthøj et al. (2018) estimate for the conversion from schizotypal disorder to schizophrenia and the 6.04 for any substance abuse that Nielsen et al. (2017) estimate for the onset of schizophrenia, both in the Danish registers.

λ_m, **the annual chance of recovering from SMI.** This parameter determines how long an SMI spell is expected to last. As it rises, SMI becomes less persistent and Treatment First's deterrence of drug use matters less, because the SMI spells a SUD can trigger are shorter on average. I set it externally to 0.066, which is an annual hazard consistent with the ten-year recovery rate of 50 percent for bipolar disorder in Åsbø et al. (2022).

ω_m, **the level of the SMI onset profile.** This parameter scales the probability of developing SMI at every age. As it rises, a SUD has a larger effect on SMI, and Treatment First reaches further than just drug use and SUD. It is calibrated internally to the share of the population with SMI, 4.82 percent, the prevalence in NSDUH 2024.

κ_m^{ue}, **how much less likely it is to regain employment with SMI.** The same as κ_d^{ue} for SMI: a household with SMI finds work at the rate $\bar{\pi}^{ue}$ divided by $\kappa_m^{ue} > 1$. As it rises, expected unemployment spells with SMI lengthen and SMI costs more. Its moment is the SMI share of the homeless, 25 percent in Gutwinski et al. (2021), since a household out of work longer is likelier to become homeless.

μ_m, **the SUD recovery multiplier for SMI.** The chain's second direction: a SUD raises the chance of SMI, and SMI makes a SUD more persistent. As it falls, recovering from a SUD is less likely, and Treatment First has more to overcome. It is set externally at 0.82 from the dual-diagnosis relapse rates in Bradizza et al. (2006).

$\kappa_{\phi m}$, **the mortality multiplier for SMI.** The chain's last link, through which SMI shortens life. As it rises, more households with SMI die, and Treatment First has more to work on, because less drug use means less SUD, less SMI and fewer deaths. It is set externally at 2.50, the middle of the standardised mortality ratios for SMI in Olfson et al. (2015).

Block 4: Recovery and treatment. Both policies work through this block: Treatment First encourages seeking treatment, and Housing First buys recovery by housing people, since recovery is likelier while housed. The first two parameters set how much recovery there is to buy, the next two how much of it treatment reaches, and the last what treatment costs the government.

$\bar{\pi}_r$**, the baseline SUD recovery rate.** This parameter is the annual chance that a housed household recovers from a SUD, the level everything else in the block scales. As it rises, disorders are expected to end more quickly on their own, and there is less for either policy to do. It is set externally at 0.025, the annual hazard implied by the ten-year remission rate in Vaillant (1995).

μ_H**, SUD recovery when homeless.** This parameter sets how much less likely recovery is in a shelter, and so how much recovery a voucher buys beyond shelter. As it falls, housing buys more recovery and Housing First is more effective in curing a SUD; near one, Housing First has little effect. It targets the effect of housing on recovery in Milby et al. (2005), who find that housing improves recovery chances by 47.3 percent.

g**, the treatment effect on recovery.** The instrument through which Treatment First reaches households that already have a SUD, which matters most in the short run. As it rises, treatment is likelier to end a SUD, and Treatment First has a larger margin to act on. It is set externally at 2.00, which is how much medication for opioid use disorder raises remission: rates near double those without it (Paul et al., 2024), and randomised trials of agonist medication against placebo find at least as large an effect on abstinence (Fudala et al., 2003).

$\bar{\tau}^u$**, the utility cost of treatment.** Take-up is a choice, and this utility cost is what makes households decline a treatment that would raise their chance of recovering. As it rises, fewer take treatment up, and Treatment First induces fewer households with a SUD to seek it. It is calibrated internally to treatment take-up: 24.5 percent of adults with a SUD received substance use treatment in the past year in NSDUH 2024.

$c_{\mathbf{treat}}$**, the public cost of a year of treatment.** This parameter is what the government pays for each household with a SUD that takes up treatment, under both policies. It ties treatment to the government's budget. Treatment First raises take-up, since entering treatment and stopping is how a household with a SUD keeps its voucher, so part of what the screen saves goes on treatment rather than on vouchers for others. As c_{treat} rises, Treatment First frees less for vouchers and π_V rises less. It is set externally to \$9,426 a year, 8.4 percent of average earnings, the cost of a residential treatment episode in Roebuck et al. (2003).

Block 5: Externalities. These two parameters decide what is an externality and how large it is.

ω_c**, the parent's weight on its child.** A parent bears the share ω_c of what its choices cost its child, in childhood and through the productivity penalty. As it rises, the parent counts more of the child's loss, so the *budget, earnings and childhood homelessness externalities* left for either policy to correct are smaller. It is identified by a counterfactual moment: drug use when the child is given no weight, against drug use at ω_c, with every other parameter unchanged.[28] The data moment holds the household fixed as well. In NSDUH 2024, an unmarried adult living with one other adult and no child was 17.5 percent more likely to have used illicit drugs in the past year than a lone parent living with one child. The lone parents are reweighted to the other group's age, sex, marital status (never married, divorced or separated, or widowed) and family income; with two members in each household, equal family income is equal income per person.[29]

[28] With $\omega_c = 0.69$, the weight on children that Nishiyama (2000) estimates and the upper end of the literature, and with b re-matched to drug use, Treatment First still raises a newborn's welfare, by 3.18 percent of lifetime consumption, against 2.57 in the benchmark. Less of that gain is an externality, however: 23 percent of it, against 76 percent in the benchmark (the last column of Table 6), because a parent with this weight counts most of what its choices cost its child, 1.64 percent of lifetime consumption against 0.19. Quantitative models in the dynastic tradition of Barro and Becker (1989) typically use lower weights (Lee and Seshadri, 2019; Abbott et al., 2019).

[29] This moment counts any past-year use of an illicit drug, a broader measure than the model's, because too few lone parents in the sample have a disorder for the comparison to give a statistically reliable measure. It is therefore used only for how much use responds, in proportion, and not for its level: the model matches it as the proportional rise in its own users.